

1. Introduction

Solarmodbus is a Home Assistant integration that communicates with hybrid inverters using **standard Modbus**, either:

- **Modbus RTU** over RS485
- **Modbus TCP** via an RS485→TCP gateway

This document provides a **generic wiring and configuration guide**, using the **Deye SG05LP1-EU-SM2-P** as a reference example. Every inverter model may expose Modbus differently, so **users must adapt wiring to their own hardware**.

2. Modbus Communication Methods

Solarmodbus supports two real Modbus paths:

A. Modbus RTU (RS485) via USB adapter

- Direct RS485 communication
- Requires a USB → RS485 adapter
- Most reliable and simplest
- Full read/write support

B. Modbus TCP via RS485→TCP gateway

- RS485 converted to native Modbus TCP (port 502)
- Works over Ethernet
- Supports multiple clients
- Full read/write support

Note: This guide does **not** use Solarman Wi-Fi sticks or port 8899 encapsulation.

3. Modbus Wiring Basics (Generic)

RS485 uses a **differential pair**:

- **A (sometimes called D+, non-inverting)**
- **B (sometimes called D-, inverting)**

These two wires must be connected to the inverter's RS485 output. Depending on the inverter model, RS485 may be exposed via:

- a **terminal block**
- a **dedicated RS485 connector**
- a **CAN/RS485 shared port**
- an **RJ45 socket** (as on our reference model)

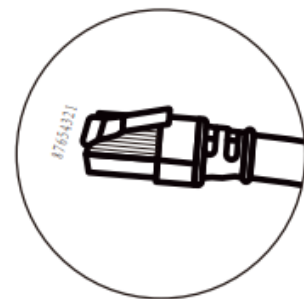
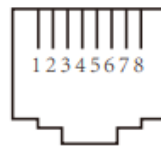
Users must check their inverter's documentation or label to identify the correct pins.

Our inverter, the **Deye SG05LP1-EU-SM2-P**, exposes Modbus RTU on an **RJ45 port** wired as follows

Definition of RJ45 Port Pin for Modbus.

No.	Modbus Pin
1	sunspe-485_B
2	sunspe-485_A
3	GND_sunspe-485
4	--
5	--
6	GND_sunspe-485
7	sunspe-485_A
8	sunspe-485_B

Modbus Port



This mean we are able to use pin 1-2 or 7-8

As we use an RJ45 cable with B-wiring we select the pair Orange-white/Orange but White-Brown/Brown is also OK.

RJ45-B color code:

Pin 1 : White/Orange
 Pin 2 : Orange
 Pin 3 : White/Green
 Pin 4 : Blue
 Pin 5 : White/Blue
 Pin 6 : Green
 Pin 7 : White/Brown
 Pin 8 : Brown

Our final choice to carry the Modbus signals is:

- **Pin 1 (White/Orange) = RS485-A**
- **Pin 2 (Orange) = RS485-B**

This is **not universal**, and this pinout is **specific to the SG05LP1-EU-SM2-P** and again check your model documentation.

Other Deye/Sunsynk models may use:

- different RJ45 pins
- a terminal block instead of RJ45
- a CAN/RS485 shared port

- or no RJ45 Modbus at all

Users must adapt wiring accordingly.

4. Connecting a USB→RS485 Adapter

Generic wiring:

Inverter RS485-A → Adapter A
Inverter RS485-B → Adapter B

Example using our RJ45 model:

White/Orange (Pin 1) → A
Orange (Pin 2) → B

Ground is optional.

Once connected, the adapter appears in Home Assistant as `/dev/ttyUSB0`.
You can check the exact name with the command:

```
ls -l /dev/ttyUSB* or ls -l /dev/ttyA*
```



5. Connecting an RS485→TCP Gateway

Generic wiring:

Inverter RS485-A → Gateway A
Inverter RS485-B → Gateway B

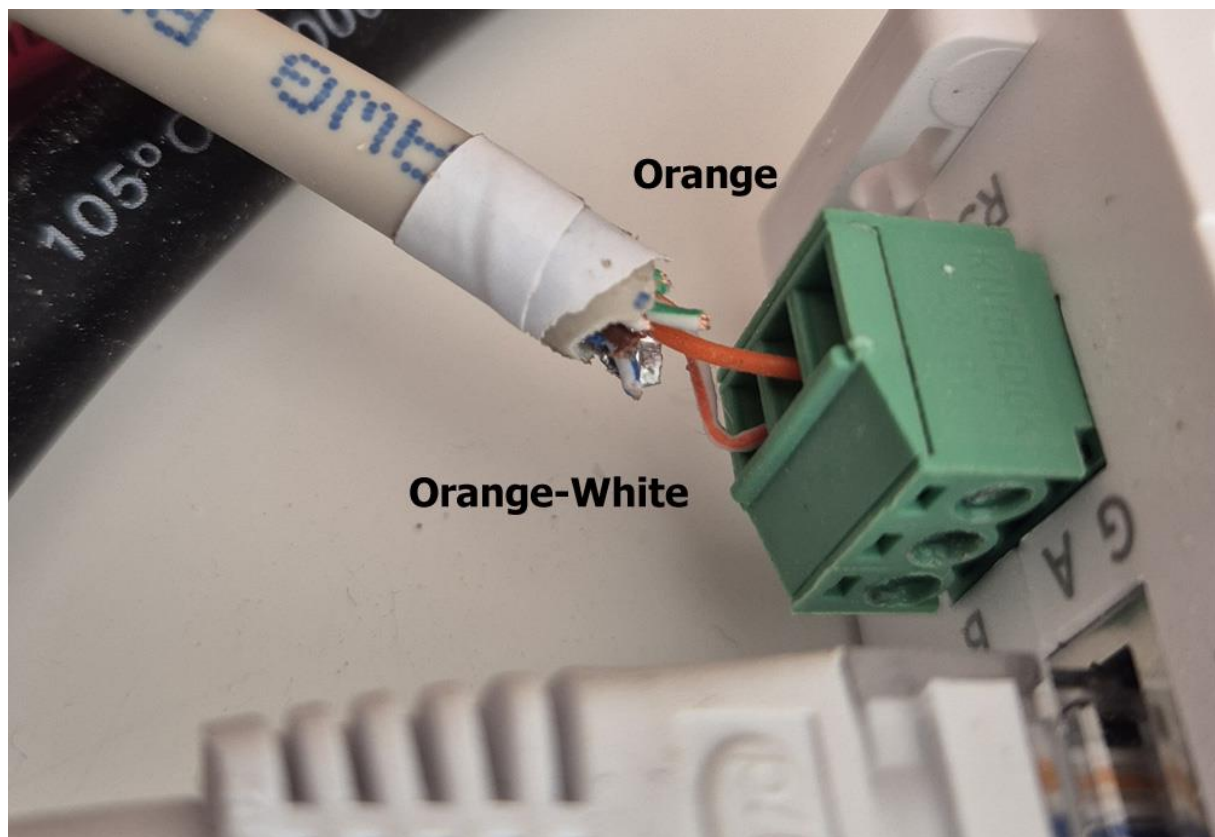
Gateway configuration:

Our reference model is the EBYTE NA111 with the following configuration:

Network parameter			
DHCP	disable	Work mode	TCP server
Local IP	192.168.20.77	Local port	502
subnet mask	255.255.255.0	Getway	192.168.20.1
Target IP	192.168.3.3	Target port	8888
MAC 0C-3D-5E-88-FD-53			
Web port	80		
DNS server	192.168.20.253		
Serial parameter			
Baud rate	9600	Data bit	8
Stop bit	1	Flow	NONE
Parity	NONE		
MODBUS parameter			
MODBUS TCP to RTU	Open	address filter	0 0:close,range:1-255
Modbus mode	disable	MODBUS polling interval	1000 Range:0-65535ms
Modbus timeout	1000 Range:0-65535ms	Modbus keep time	5 Range:0-255s
		Mosbus instructions add clear spare space 49 01 03 00 00 00 0A X	

Don't worry about Target IP, Target port you can't change these values and they are not used.

Gateway wiring detail:

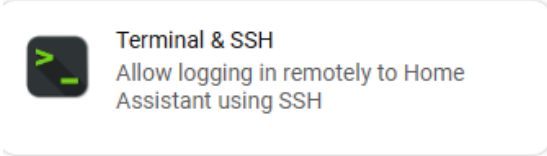


6. How to install on HAOS

Install the SSH App:

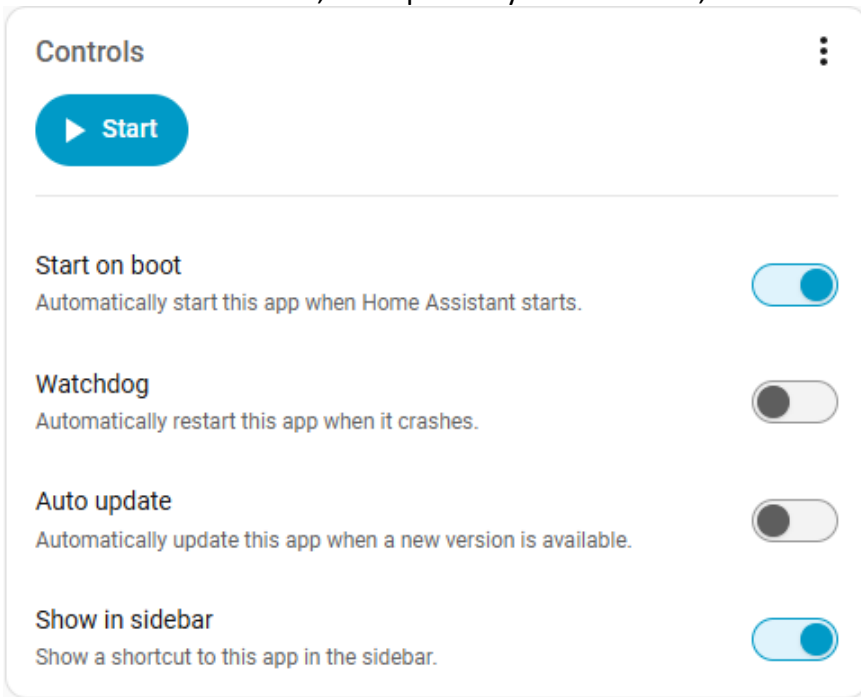
Settings → Apps → Install App, **search and install:** Terminal & SSH

Official apps



Configure it as follows:

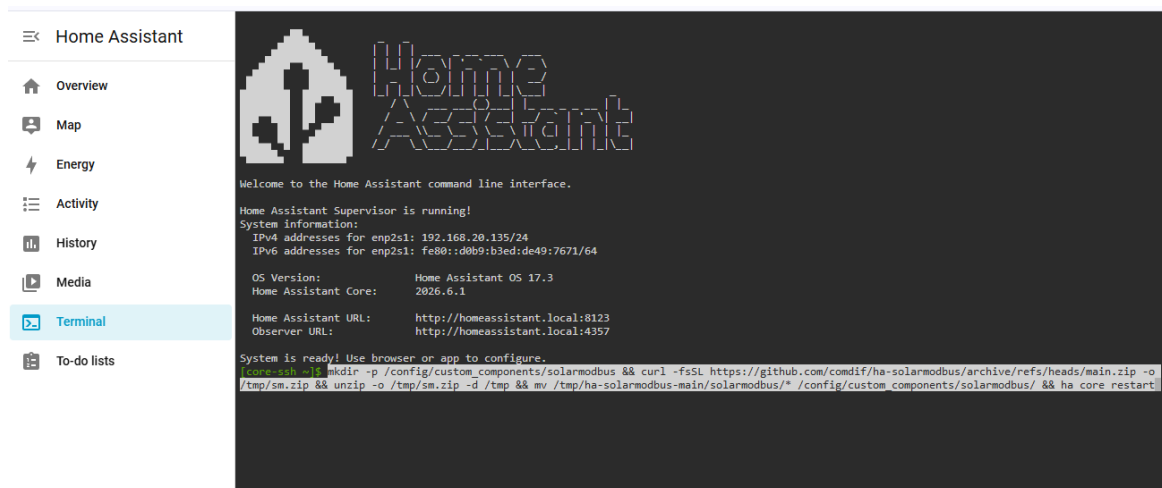
Turn on Show in sidebar, and optionally Start on boot, then start App.



On the left menu sidebar open Terminal Copy & paste your one-liner

Copy this block and paste it:

```
mkdir -p /config/custom_components/solarmodbus && curl -fsSL https://github.com/comdif/ha-solarmodbus/archive/refs/heads/main.zip -o /tmp/sm.zip && unzip -o /tmp/sm.zip -d /tmp && mv /tmp/ha-solarmodbus-main/solarmodbus/* /config/custom_components/solarmodbus/ && ha core restart
```



- Press Enter
- The integration installs itself and HA will restart.

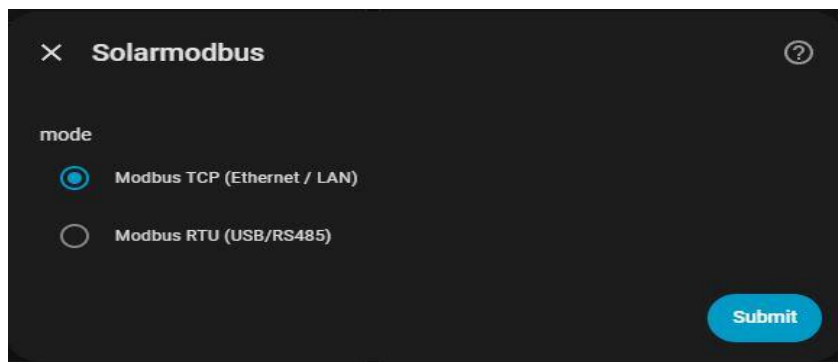
7. How to install on HA Venv or others

Just use your favorite SFTP client, copy the solarmodbus folder into your custom_components directory and restart Home assistant as usual.

8. Home Assistant Setup

Solarmodbus is configured **entirely through the UI**:

1. Go to **Settings** → **Devices & Services**
2. Click **Add Integration**
3. Search for **Solarmodbus**



4. Choose your mode: Enter Serial port (for USB) or I P address (for TCP)

TCP (gateway)

×

Solarmodbus

?

host*

192.168.20.77

port*

502

slave_id*

1

model *

deye_hybrid

Submit

Or USB (RTU)

×

Solarmodbus

?

help_text

☒

Serial device path. Common values: /dev/ttyUSB0, /dev/ttyUSB1, /dev/ttyACM0. Must be verified on the system.

serial_port*

/dev/ttyUSB0

slave_id*

1

model *

deye_hybrid

Submit

5. Select your inverter model definition

For our model **Deye SG05LP1-EU-SM2-P** our selection is **deye_hybrid**

6. Submit — entities appear automatically

No YAML. No manual modbus definitions. No register mapping. Everything is handled by the integration.